while not offering any extra features. These drives are primarily meant for small offices, where people would want to connect them to different machines for backups.

We received four drives in this category: the BenQ EW162I, Lite-On SHW-1635SU, Plextor PX-716UF, and the Sony DRX-800UL.

Features

Most external DVD-Writers are in fact internal writers encased in a rugged casing, and with external connectivity. Thus, the Lite-On SHW-1635SU is the same as the Lite-On SHW-1635S, the Plextor PX-716UF is the PX-716A, and the Sony DRX-800UL is the DRX-800A. The internal versions of the first two of these—the SHW-1635S and the PX-716A—have been featured in the previous section. The external versions, naturally, have many of the same features.

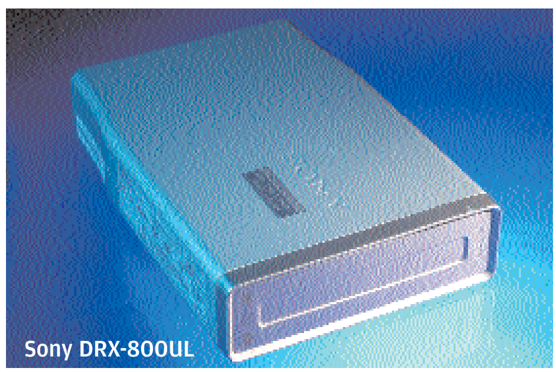
Traditionally, external DVD-Writers have been a generation behind the internal writers, and were slower. Not any more. The external drives featured in this test boasted of the highest write speeds of 16x for DVD-R and DVD+R, just like the internal drives. The exception to the rule was the BenQ EW162I, which has been around for a long time. It supported only 4x write speeds for DVD+RW, as against 8x supported by the other drives. It also lacked support for the DVD-R DL format.

Interfaces

The BenQ and Lite-On drives had only the USB 2.0 interface, while the Plextor and Sony featured both the USB 2.0 (480 Mbps) as well as the FireWire (400 Mbps) interface. The drives are backward-compatible with USB 1.1. All the drives shipped with the cables for the supported standards, as well as the power adapters.

Physical Considerations

All the drives had rugged build quality, and we couldn't decide which was better. The Sony and Plextor drives sported killer looks. The BenQ's casing had perforations to aid ventilation—this is very helpful in Indian conditions.



Sony DRX-800UL

All the drives except the Lite-On shipped with stands to help keep them upright—a vertically-placed drive helps save valuable PC desk space.

EZ-DUB

A unique feature on the Lite-On drive was the EZ-DUB function. The drive ships with EZ-DUB software from Ulead, which is a CD/DVD burning and data backup software. After you install it, the software detects whether you have an EZ-DUB compatible drive, and sits quietly in the System Tray. Most of the EZ-DUB functionality comes from the software, but the drive, too, has two extra buttons on the top that make it even easier to burn DVDs (or CDs).

The 'File' button launches the EZ-DUB software, and a Wizard helps you drag and drop files you wish to write to disc, to the EZ-DUB main window. When you are done selecting the files, you press 'File' again to burn the compilation. The 'Dub' button is for making a 1:1 disc copy.

Next-Gen storage: Blu-ray vs. HD-DVD?

Even as we progress from the *meagre* 700 MB storage space of the CD to the 4.7 GB offered by DVDs (or even the 8.5 GB offered by dual-layer DVDs), we are left wondering if that much capacity is enough for our ever-growing digital real-estate needs. And here's where the next-gen standards of storage media—Sony's Blu-ray and Toshiba's HD-DVD—come into play.

Both these formats support considerably higher data storage capacities—Blu-ray discs can store 25 GB on the single layer and 50 GB on the dual-layer version, and HD-DVD has proven that it can store 15, 30 and 45 GB of data in the single, dual and triple-layer versions respectively. TDK has reportedly produced a 100 GB quad-layer Blu-ray disc, which puts it miles ahead of HD-DVD where sheer data volume is concerned. The upgradeability path to these optical drives requires the consumer to purchase drives supporting these standards, as neither of these standards is backward-compatible (similar to the case of DVD a few years ago). This is because both these standards deploy a lower-wavelength, 405 nm laser. This falls in the violet-blue part of the electromagnetic spectrum—hence the name Blu-ray. A different and more expensive laser head is hence called for. Also, the lower wavelength of the laser allows data to be written at a higher resolution, enabling a much higher amount of data to be packed onto the same surface area as a DVD or CD.

The media types of Blu-ray and HD-DVD are also different from that of DVD. While HD-DVD uses a specification that is much closer to that of DVD, a 0.6 mm thick base and a 0.6 mm thick protective layer, Blu-ray uses a thicker 1.1 mm base but a much thinner, 0.1 mm protective layer. As far as data transfer rates go, Blu-ray is again the front-runner, being able to achieve a speed of 12x (400 Mbps) at the same rotational speeds at which a HD-DVD drive will be able to achieve 9x. Such speeds might not matter much today, but to accommodate the needs for higher data throughput in the future, this is sure to make a difference.

The Blu-ray "camp" includes Sony, TDK, Dell, Hitachi, Apple, HP, Philips and Samsung, and consumer electronics giants such as Panasonic, Samsung, Sharp, Pioneer and LG Electronics—as also game makers EA and entertainment companies Twentieth Century Fox, Vivendi Universal, Walt Disney and Sony Pictures. HD-DVD is backed by NEC, Sanyo, and heavyweights such as Intel and Microsoft, as well as entertainment companies such as HBO, New Line Cinema, Universal Studios Home Entertainment and Warner Home Video. In late December 2005, HP announced that it would stop taking sides, and support both formats.

Talks for a hybrid disc technology, which were in the news a while ago, are in the cold as of now, and hence backward compatibility of these media seem out of the question. At the end of the day, the consumer is left to decide which side to choose. On one side is Blu-ray with its bevy of supporters, along with its higher capacity and data transfer speeds. On the other side is HD-DVD with its lower price tag, and the vital backing of the biggest names in the computing world—Intel and Microsoft—which guarantees it native support in future computers and operating systems such as Vista. When these drives begin shipping in 2006, the consumer will face a daunting task—which way to upgrade?